

NIFTY 50 Market Regime Forecasting

Project report - machine-learning models that forecast the market 'regime' for intraday, weekly-expiry and monthly-expiry trading. Validated honestly (no cheating), with simple explanations.

1. What this project does (in one line)

It looks at NIFTY 50 price behaviour and forecasts the upcoming market REGIME - the 'mood' of the market - so an options/expiry trader knows what to expect. We built one model for each time-frame: intraday, weekly expiry, and monthly expiry.

2. The three models (names + simple meaning)

- **Intraday Volatility-Regime Model (HMM + XGBoost hybrid):** Predicts whether the next few hours will be CALM or VOLATILE (big moves). Use: calm -> sell option premium; volatile -> buy / hedge.
- **Weekly Direction Model (HMM + XGBoost hybrid):** Predicts the market DIRECTION over the next 5 trading days - BULL (up), BEAR (down) or SIDEWAYS (flat) - to position for the weekly expiry.
- **Monthly Volatility Model (Volatility-Persistence rule):** Predicts whether the next ~21 trading days will be CALM or VOLATILE. A simple, robust rule (volatility tends to persist) beat the complex models here.

3. Results - how accurate are they?

'Hit ratio' = how often the model is right. 'Baseline' = a naive guess (always pick the most common answer). 'High-confidence' = accuracy when we only act on the model's most confident calls.

Model	Hit ratio	Baseline	High-confidence
Intraday (volatility)	64.6%	53.6%	78%
Weekly (direction)	58.0%	47.5%	67%
Monthly (volatility)	73.7%	56.4%	85%

Every model beats its baseline by +10 to +17 points. Trading only the confident calls raises the hit ratio further (e.g. intraday 64.6% -> 78%).



4. How we made sure the results are HONEST (no cheating)

- **No look-ahead:** We always train the model on the PAST and test it on UNSEEN FUTURE data (walk-forward). It never sees the answers in advance.
- **Leakage test (shuffle):** We scrambled the training answers; the model's score collapsed to a random guess - proving it was not secretly using future information.
- **Overfitting check:** We checked the gap between training and testing accuracy and simplified the models so they GENERALISE instead of memorising.
- **What did NOT work:** We tested India VIX and cross-asset data (Bank Nifty, USD-INR, crude) - they did not help, so we honestly dropped them.

5. What was done today (step by step)

- Set up the tools (installed Python + the ML libraries).
- Built and tested the three forecasting models for NIFTY 50.
- Found that intraday/monthly DIRECTION is unpredictable - so we predict VOLATILITY there instead.
- Found the best model differs by time-frame: hybrid (HMM+XGBoost) for intraday & weekly; a simple persistence rule for monthly.
- Combined two models (HMM + XGBoost) into a stronger hybrid.
- Checked every model for cheating (leakage) and memorising (overfitting), and fixed them.
- Built a dashboard page to visualise the results and pushed everything to GitHub.

6. Market terms (easy meanings)

Regime: The market's current 'mood' - trending up (bull), down (bear), flat (sideways), or calm vs volatile.

Volatility: How much the price swings. High volatility = big, fast moves.

Bull / Bear / Sideways: Going up / going down / going nowhere.

Expiry: The day options contracts settle (weekly = ~5 days, monthly = ~21 days).

Hit ratio / Accuracy: How often the prediction is correct.

Baseline: A naive benchmark - always guess the most common outcome.

Confidence: How sure the model is. We trade only high-confidence calls.

India VIX: The market 'fear gauge' - rises when traders expect big moves.

Option premium: The price of an option. Sellers profit in calm markets; buyers in volatile ones.

Walk-forward / Out-of-sample: Testing on future data the model never trained on - the honest way.

Look-ahead / Leakage: Accidentally letting the model peek at the future - makes fake-good results.

Overfitting: When a model memorises the past instead of learning - looks good, fails live.

7. Models used (technical names)

- **Gaussian HMM (Hidden Markov Model):** A statistical model that finds hidden market 'states' and how they switch over time.
- **XGBoost:** A powerful 'gradient-boosted decision tree' model - the industry standard for this kind of data.
- **Hybrid (HMM + XGBoost):** We feed the HMM's regime read into XGBoost - the two together beat either

alone.

8. Status & next step

All three models are validated, leak-free and overfitting-checked - ready for PAPER TRADING (test with no real money first). 64% intraday accuracy is a genuinely good edge (most real edges are 55-60%). Next: refine the labels and add live data, then go live small.